

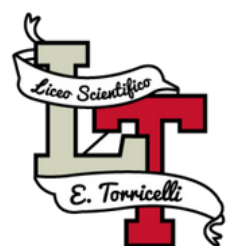


TORRICELLI

RACING

DESIGN & ENGINEERING

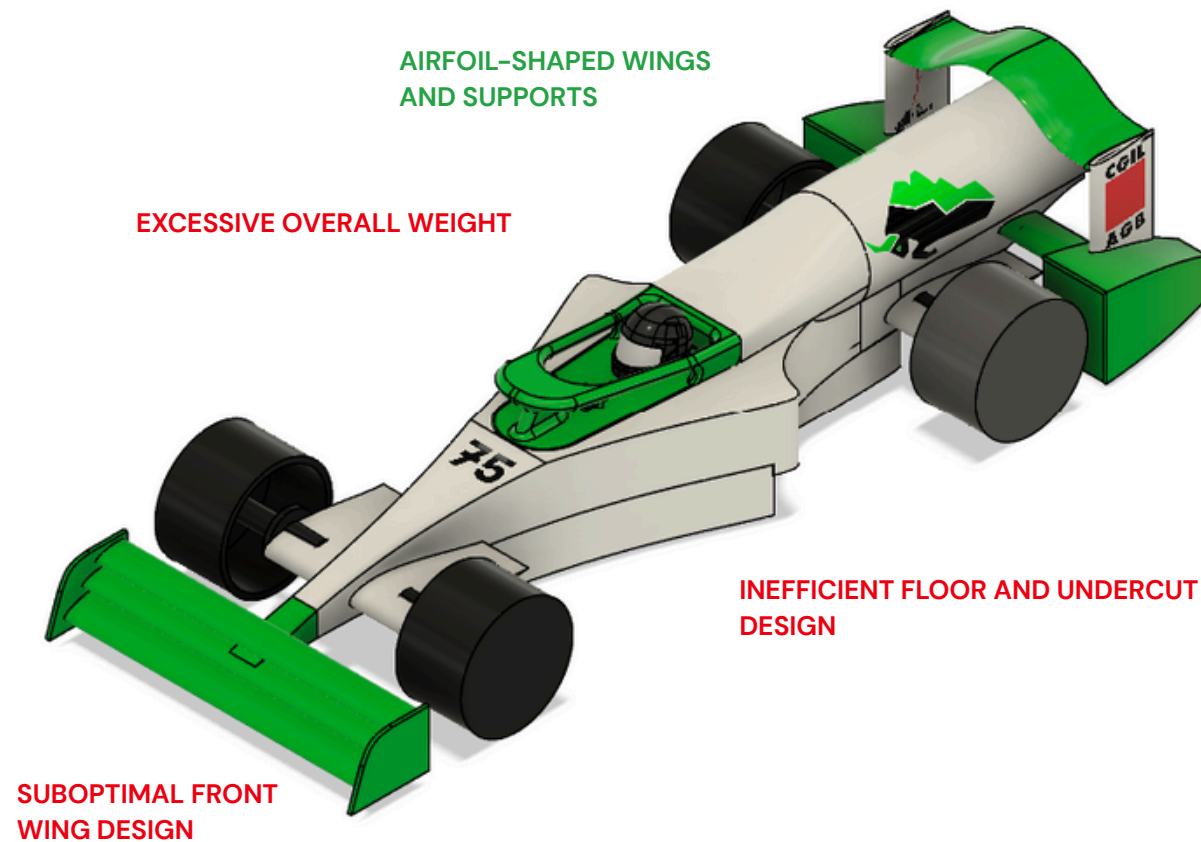
PORTFOLIO



QUALIFIERS' CAR ANALYSIS

After completing the regional qualifiers, we carefully analyzed the performance of our car. Although it showcased some good aerodynamic ideas — like airfoil-shaped wings and airflow management devices — several weaknesses were identified, highlighting the need for a major redesign.

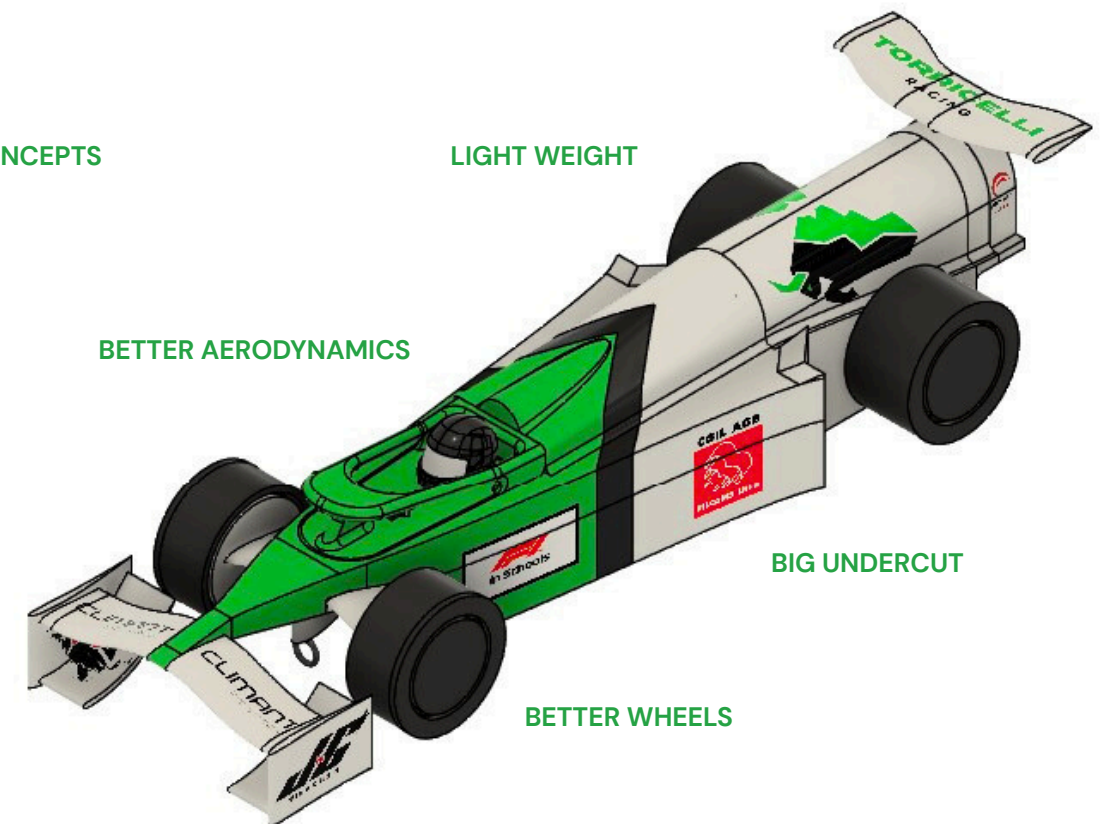
QUALIFIERS' CAR



EXPLANATION: **STRENGTHS** / **WEAKNESSES**

DECENT AERODYNAMIC CONCEPTS
(E.G., AIRFLOW CLEANERS)

VS



REGIONAL FINALS' CAR

MAIN GOALS

Based on this analysis, we decided to completely redesign the car while retaining some successful concepts. Our **main goals** were:

- **Reducing** the **weight** down to the minimum limit by redimensioning some shapes (like minimizing the length of the chamber) and changing materials to some better performant ones
- Redesigning the **front wing** for improved **airflow management** and **reduced drag** and removing unnecessary parts
- **Optimizing** the **floor** and **undercut** for a **cleaner** and **faster air passage**
- Introducing **bearing-based wheel assemblies** to **minimize friction** and **rotational resistance**

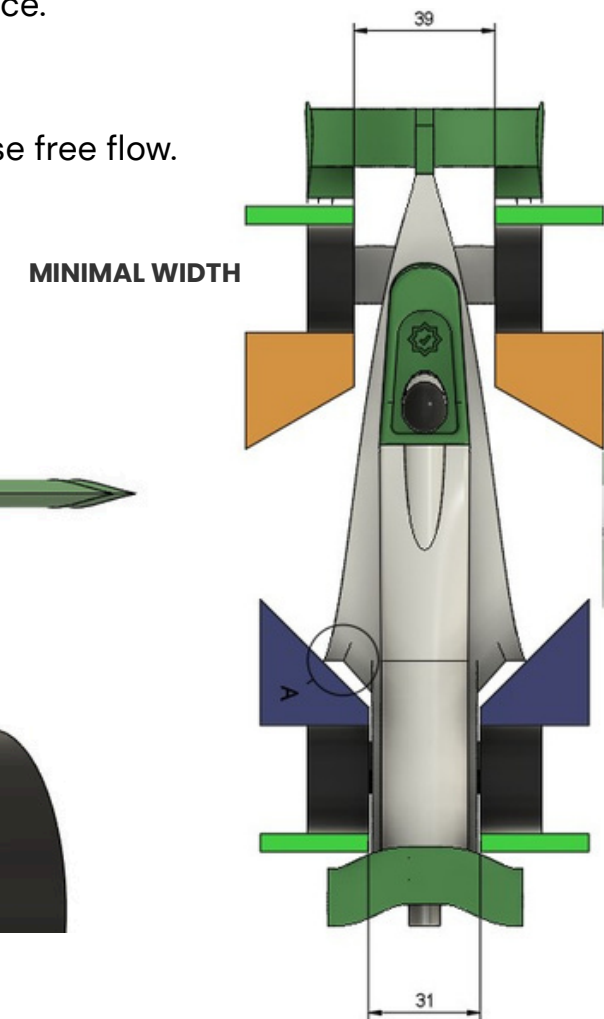
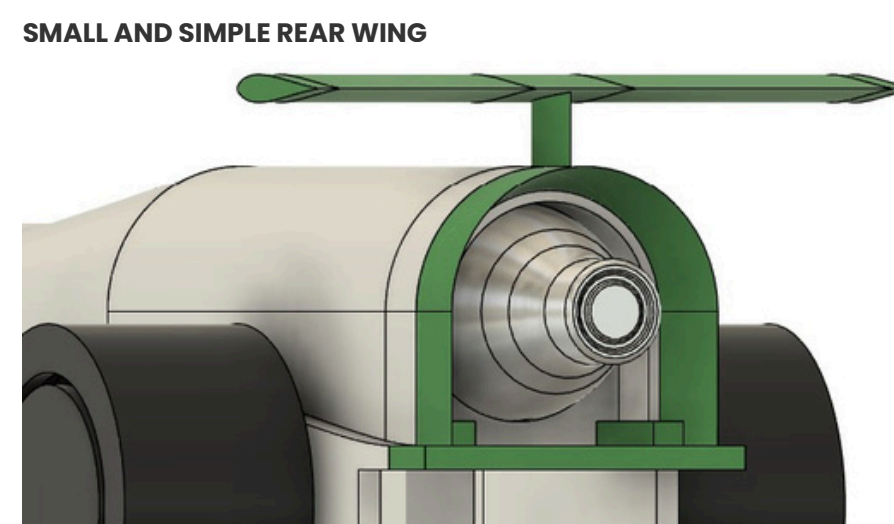
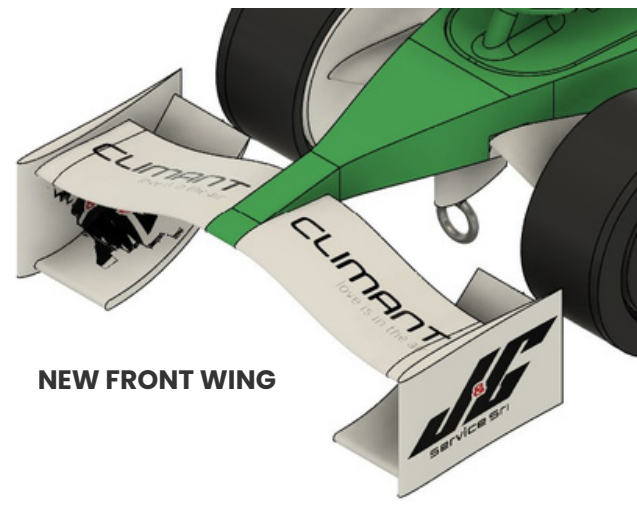
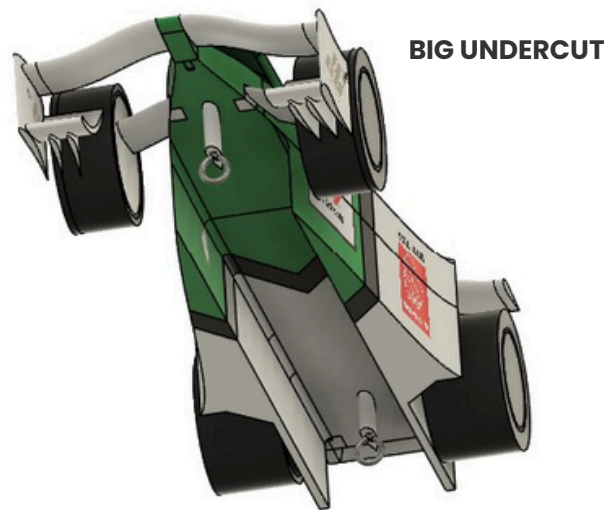
The regulatory updates allowed us to innovate further, particularly in the wheel and support structure design, pushing us to rethink every element with performance optimization in mind.

KEY DESIGN OBJECTIVES

At the start of the redesign phase, we set two main design priorities: maximizing underbody airflow (big undercut) and minimizing the car's frontal surface area (narrow wheel axis width). These choices aimed to reduce aerodynamic drag while maintaining stability and regulatory compliance.

KEY OBJECTIVES

- **Big undercut:** create the largest possible free airflow path under the car to reduce ground friction and air turbulence and maximise free flow.
- **Minimal width:** design a narrow wheel axis width and small wheels to lower drag and minimize frontal area.
- **New front wing:** redirect airflow efficiently around the front wheels without blocking the underbody airflow.
- **Lightweight components:** replace or redesign heavy parts to achieve minimum weight.
- **Simple and small rear wing:** keep the rear wing lightweight and aerodynamically clean to avoid unnecessary drag



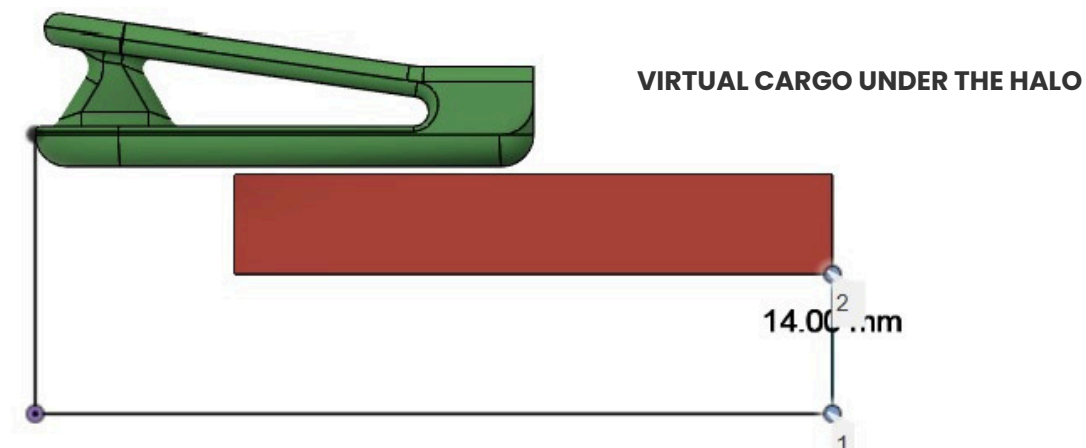
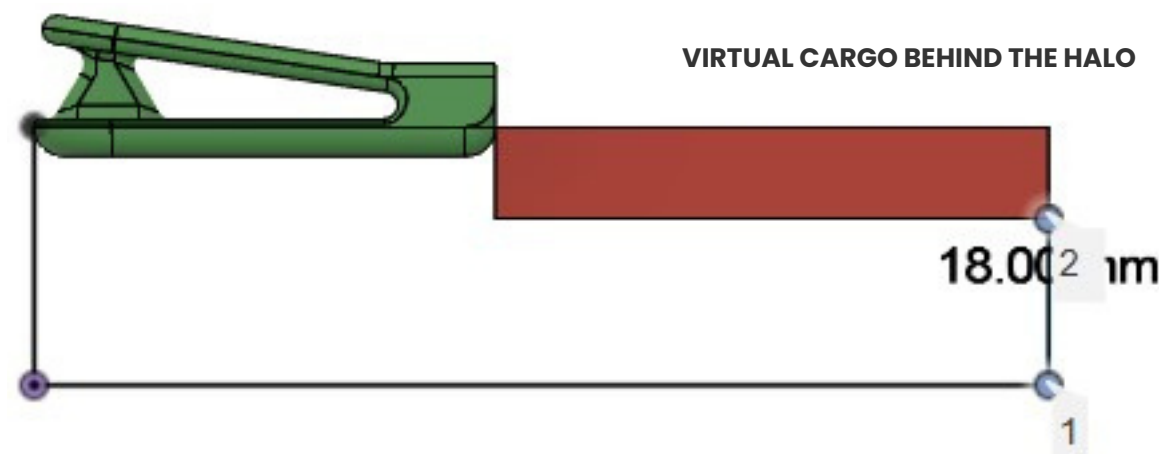
SKETCHING PHASE

In the early sketching stage, we explored two possible layouts for the Virtual Cargo (mandatory volume):

- **Virtual Cargo placed under the Halo**
- **Virtual Cargo placed behind the Halo**

We chose the second option as it allowed a higher and more continuous undercut, better supporting our key aerodynamic concept.

Further design decisions — such as the placement of sidepods, deflectors and halo connections — were all influenced by our goal to maintain a clean, unrestricted airflow underneath the car.



FRONT WING & SUPPORT DEVELOPMENT

REDESIGN MOTIVATION

The front wing used in the qualifiers was aerodynamically inefficient and heavy. To improve performance, we chose to completely redesign the wing, focusing on **drag reduction**, regulatory compliance and structural simplicity.

DESIGN CHOICES AND FEATURES

- A **single-vane** wing was selected for simplicity and reduced frontal impact with air.
- It is attached to a short, **thin nose cone**, minimizing drag and maximizing clean airflow.
- Endplates were placed in front of the wheels, designed to **redirect air** both around and over them, while complying with rule T8.7.
- The first version of the endplates was too heavy and forced an excessively narrow wing. We redesigned them to be slimmer and more effective.
- The wing is mounted **high** above the ground to eliminate the need for extra connectors (which would increase drag) and is slightly **bent** downwards to meet rule T8.5.2.
- A **neutral airfoil shape** was used, avoiding both lift and downforce to maintain clean airflow.

MANUFACTURING AND ASSEMBLY

- The entire wing and nose assembly is printed as a **single piece**, ensuring precision and reducing weight.
- A custom mounting point was designed to connect the front wing directly to the car body, offering rigidity with minimal aerodynamic disruption.

PERFORMANCE IMPROVEMENT

Compared to the first version, our final design showed a **19%** reduction in **drag** based on CFD analysis — confirming the aerodynamic benefits of our approach.

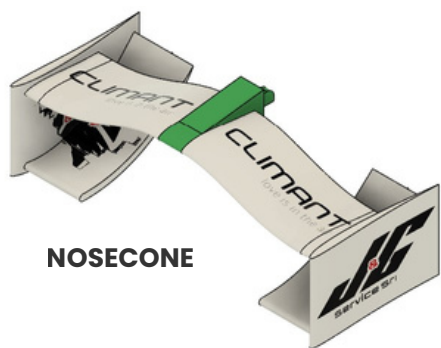
DEVELOPMENT ITERATIONS & CFD TESTING

To address the aerodynamic issues of the initial designs, we tested over **ten** different front wing specifications — many with small adjustments, but a few with major conceptual changes.

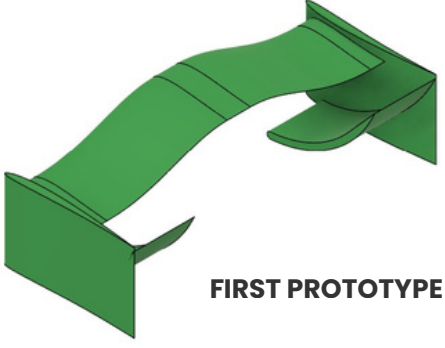
- We first modified the angle of the **endplate's airfoil**, making it **less steep**. CFD results improved slightly, but the problem wasn't fully solved.
- Next, we tested a **double-vane** endplate, which reduced drag further, but still didn't meet our expectations.
- Finally, we created a more unconventional solution: we added **cambered airfoils** underneath the endplates to redirect air **sideways**. This approach showed a significant improvement in drag reduction and was integrated into the **final design**.

Each version was simulated using Autodesk CFD and evaluated with visual flow maps and pressure values.

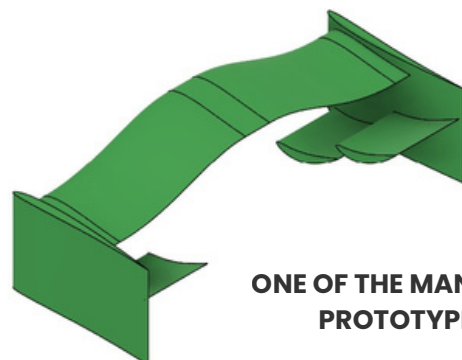
FEATURE	FINAL DESIGN
WING TYPE	SINGLE VANE
AIRFOIL PROFILE	NEUTRAL
ENDPLATE POSITIONING	FRONT OF WHEELS
NOSE CONE	THIN AND SHORT
ASSEMBLY	PRINTED AS SINGLE BLOCK
DRAG REDUCTION (CFD)	~19% VS. FIRST DESIGN
RULE COMPLIANCE	T8.5.2, T8.7



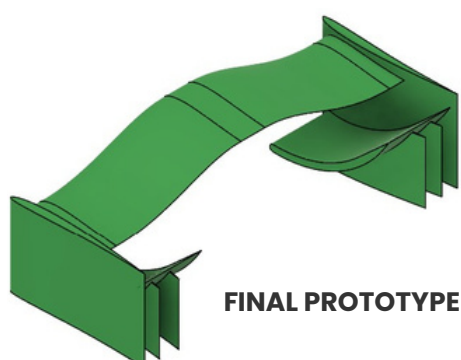
NOSECONE



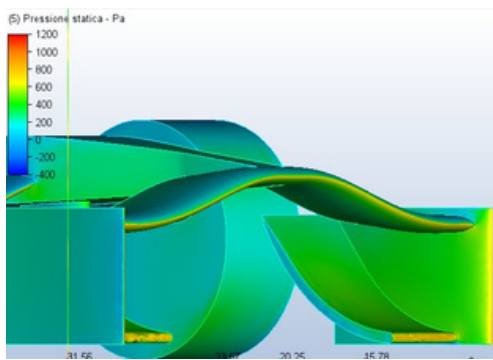
FIRST PROTOTYPE



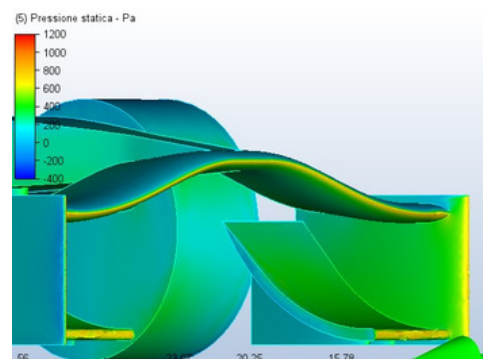
ONE OF THE MANY PROTOTYPES



FINAL PROTOTYPE



FIRST PROTOTYPE'S CFD



FINAL PROTOTYPE'S CFD

REAR WING & MOUNTING SYSTEM

REDESIGN MOTIVATION

The rear wing from the qualifiers was overly **complex** and difficult to assemble. Since the rear wing generates little aerodynamic benefit and primarily **adds drag**, our goal was to **simplify** it while maintaining stability and aesthetic integration.

DESIGN EVOLUTION

We removed the previous connection system, which attached the wing to rear deflectors, as it was unnecessarily complicated and tested two new mounting solutions:

- a. A wing that **twists** and connects directly to the chamber walls.
- b. A simplified add-on behind the chamber, with a small **airfoil-shaped support** connecting to the center of the wing.

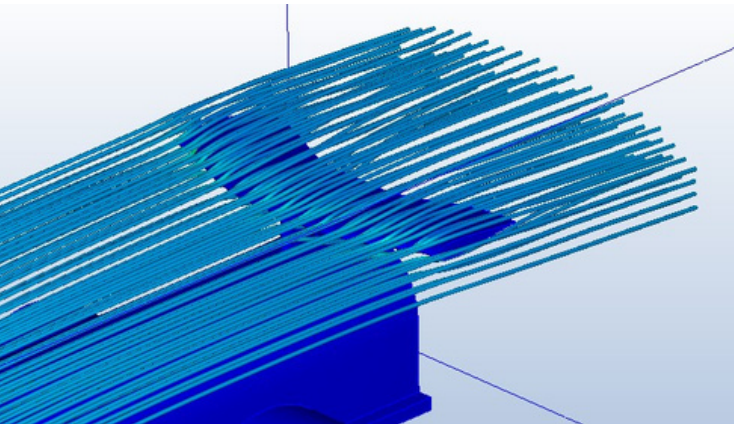
After comparison, we chose the second design due to its lower surface impact, improved stability, and ease of manufacturing.

KEY FEATURES

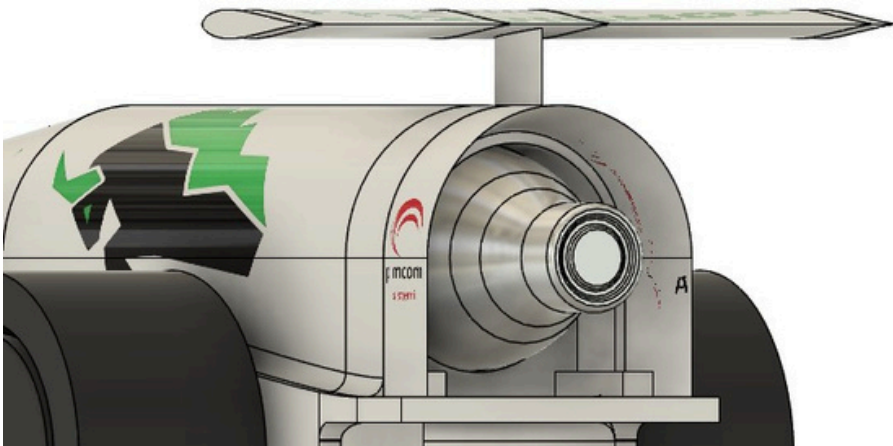
- Compact and **clean shape** with minimal drag footprint.
- **Shortened rear section** of the car to compensate for the longer wheelbase caused by halo and virtual cargo placement.
- Final wing has a **curved profile**, which better integrates with the car's structure and makes the wing be compliant with both rule T7.9 and T9.4.1

OUTCOME

The new rear wing is aerodynamically cleaner, structurally simpler, and much easier to mount — improving both performance and assembly workflow.



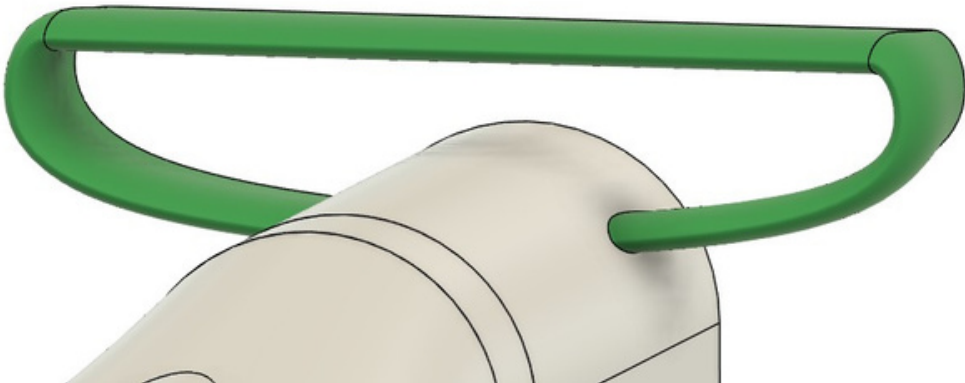
REAR WING CFD ANALYSIS: CLEAR FLOW WITHOUT TURBOLENCES



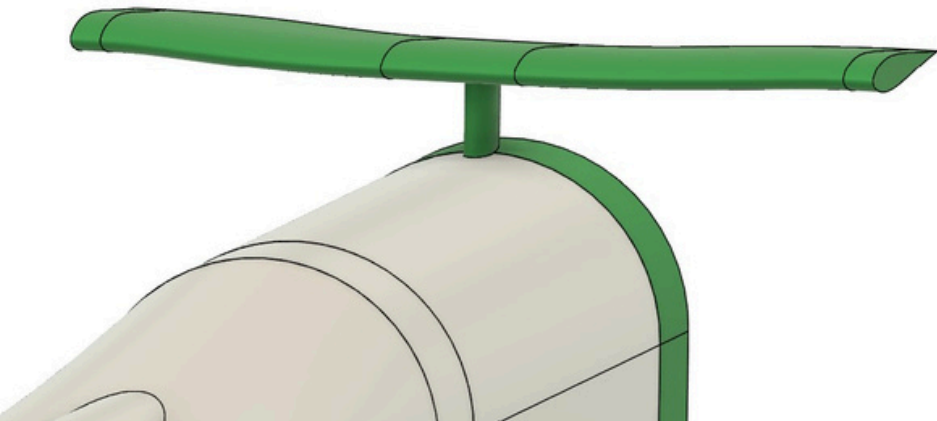
REAR VIEW OF THE REAR WING

FEATURE	FINAL DESIGN
SUPPORT TYPE	ADD-ON WITH CENTER STRUT
CONNECTION POINT	REAR OF CHAMBER
ASSEMBLY COMPLEXITY	VERY LOW
DRAG IMPACT	SIGNIFICANTLY REDUCED

TWISTED WING



MOUNTED WING



DESIGN OBJECTIVES

One of the key goals in our redesign was to maximize **free airflow** beneath the car by creating a wide and straight undercut. At the same time, we aimed to minimize turbulence and ensure structural simplicity.

ELEMENT POSITIONING AND INITIAL DECISIONS:

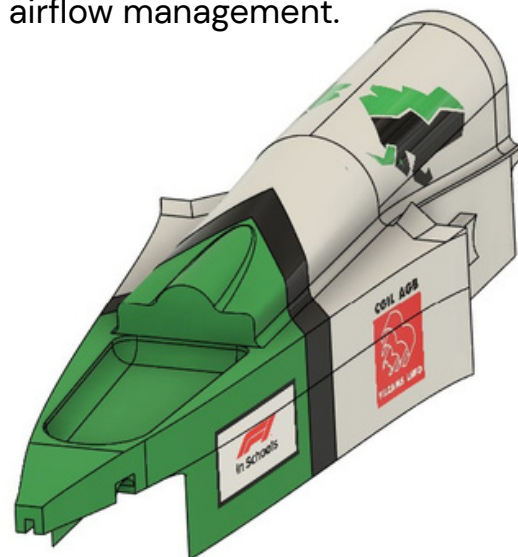
During the sketching phase, we compared two possible layouts for the Virtual Cargo: **below** or **behind** the Halo. We chose the second option as it allowed a **taller** and more continuous undercut, improving airflow performance while still complying with all regulations.

UNDERCUT & TUNNEL DEVELOPMENT

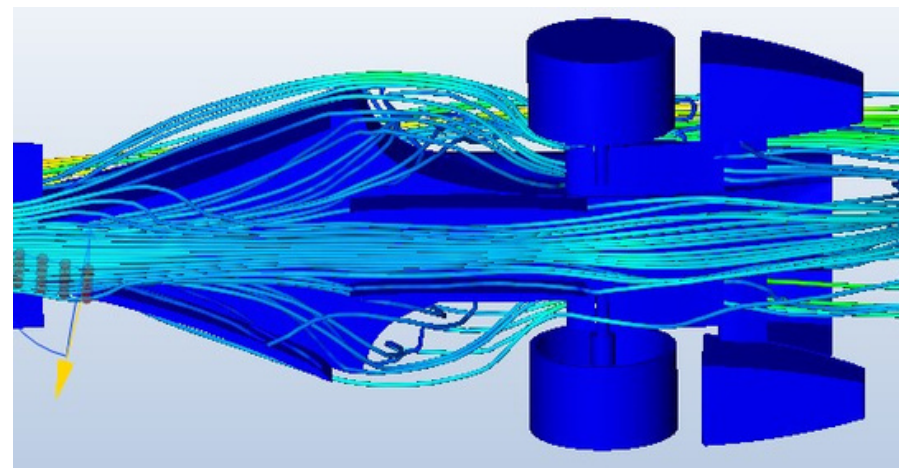
- The undercut area was redesigned to be **wider** and more stable than in the qualifiers' car.
- In the previous design, the use of deflectors under the car caused unwanted turbulence. We replaced them with tunnel **walls** that keep the airflow attached and laminar.
- These walls maintain the **same width** for their entire length to avoid flow separation.
- A simple vertical connection links the chamber to the tunnel roof, avoiding unnecessary curves and aiding clean flow.

SIDEPODS & HALO-CHAMBER CONNECTION

- We added **sidepods** raised **2 mm** off the ground to redirect air around the rear wheels.
- The top of the Halo connects to the chamber with a clean **lofted transition**. A small **cylindrical-shaped** part was added to the loft in order to clean the turbulences generated by the helmet. The bottom half forms vertical walls connecting directly to the tunnel roof, simplifying geometry and aiding airflow management.

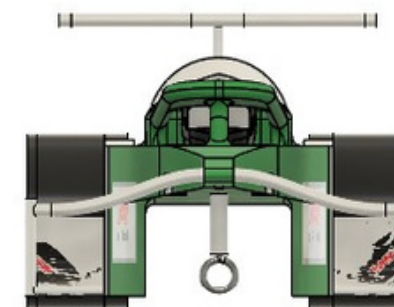


SIDEPOD & HALO-CHAMBER CONNECTION

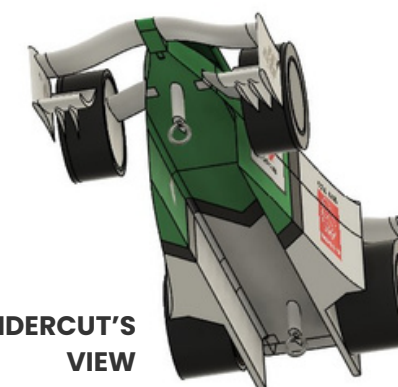


QUALIFICATIONS' CAR'S UNDERCUT FLOW: PART OF THE FLOW IS DIVERTED Laterally (COANDA EFFECT & VENTURI EFFECT) WHICH GENERATES UNWANTED TURBULENCE

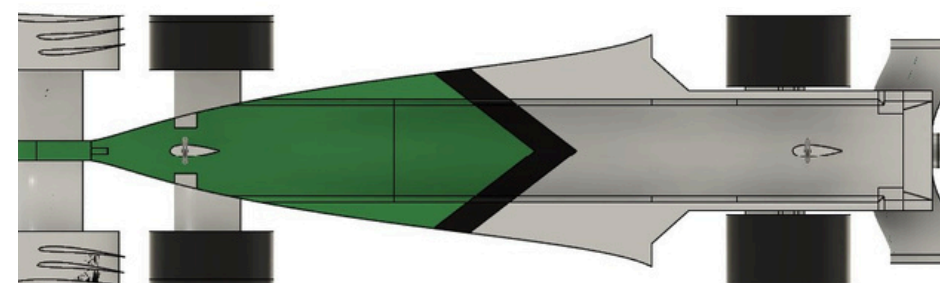
FEATURE	DESIGN CHOICE
VIRTUAL CARGO	BEHIND HALO
UNDERCUT SHAPE	CONTINUOUS, WIDE
TUNNEL WALLS	CONSTANT-WIDTH, STRAIGHT
SIDEPODS HEIGHT	2 MM FROM GROUND
HALO-CHAMBER LINK	TOP: LOFTED / BOTTOM: VERTICAL



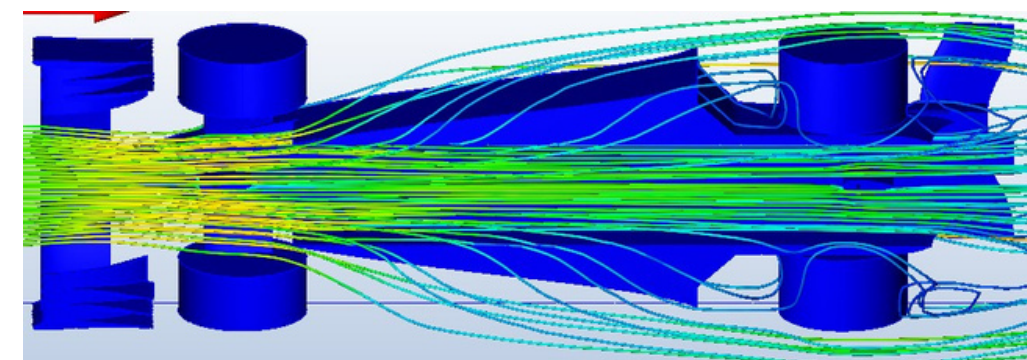
UNDERCUT'S FRONT VIEW



UNDERCUT'S VIEW



UNDERCUT'S BOTTOM VIEW



FINALS' CAR'S UNDERCUT FLOW: THE FLOW IS STRAIGHT WITHOUT UNWANTED DEVIATIONS

WHEELS & SUPPORT STRUCTURE REDESIGN

With the updated regulations allowing **bearing-based** wheel assemblies, we aimed to reduce friction between the wheels and supports, minimize drag and decrease the frontal impact surface by designing **smaller** parts.

WHEELS

Compared to the qualifiers' car (17.7 mm wide wheels), the new design features **13.1 mm** front wheels and **17.1 mm** rear wheels. The central part of each wheel is **fixed** to reduce turbulence caused by the rotating wheel. Due to rule constraints (**Pirelli sticker** placement), we slightly changed our ideas and added a smooth **side surface** on the wheel for outer stickers, while inner stickers are applied to the central hub. The use of smaller wheels significantly reduced air resistance and improved aerodynamic efficiency, especially at the front.

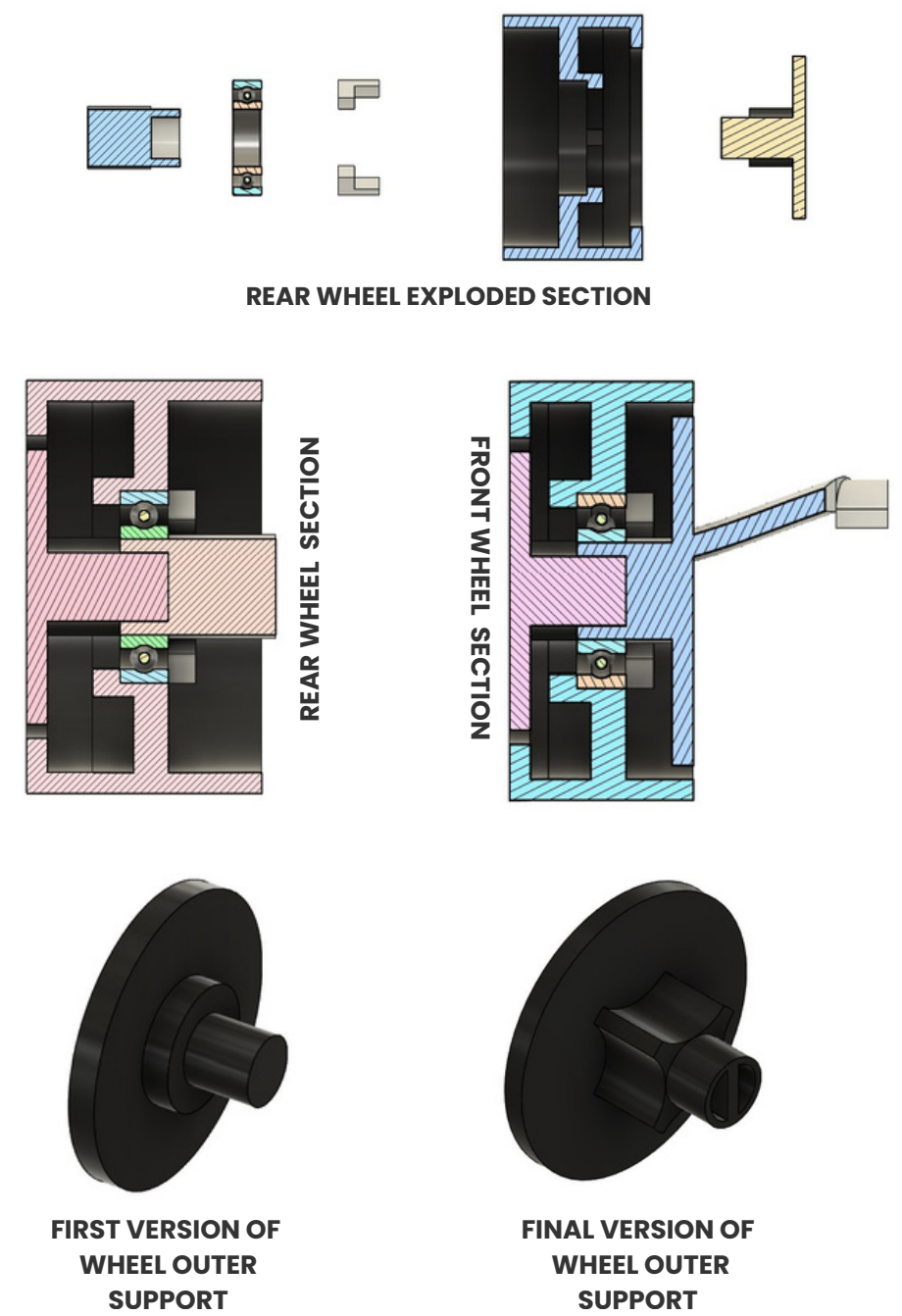


SUPPORT STRUCTURE

We designed a custom **three-part** support system. The front connection features an **airfoil-shaped** support for better airflow, while the rear uses a **lighter**, non-aerodynamic connector — as air is diverted away from that area.

Bearings are embedded directly into the support structure and wheels using a dual-axle connection system. This configuration reduces rolling resistance and **eliminates friction** between moving parts. Despite the added complexity, no printing or fitting issues were encountered. The new design also reduced overall wheel system weight by approximately **1.5 grams** — and, more importantly, greatly improved **aerodynamic performance** by reducing drag-heavy contact surfaces.

COMPONENT	QUALIFIERS	FINALS
FRONT WHEEL WIDTH	17.7 MM	13.1 MM
REAR WHEEL WIDTH	17.7 MM	17.1 MM
TOTAL WEIGHT SAVED	-	≈1.5 G



WEIGHT REDUCTION STRATEGY & FINAL MASS

INITIAL CHALLENGE

Our first complete design was over **10 grams above** the weight limit (qualifiers car: ~95g without CO₂ canister). Since the F1 in Schools block has low density, we realized that major reductions had to come from **3D-printed components**. The bearings (8g total) were non-negotiable, wings were already small, so we could mostly trim the **wheels**.

REDUCTION PROCESS

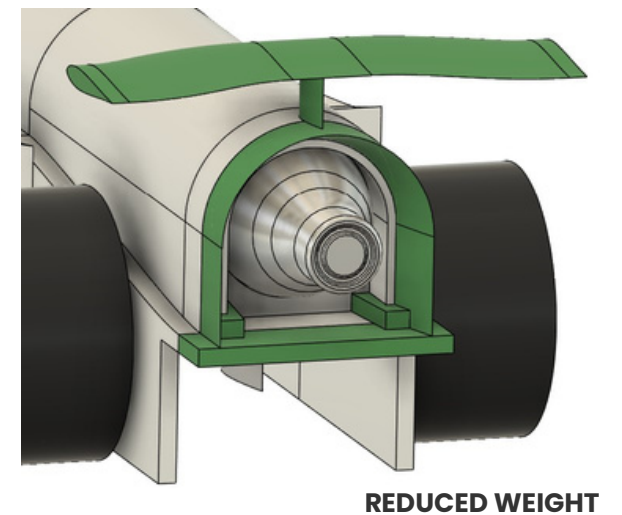
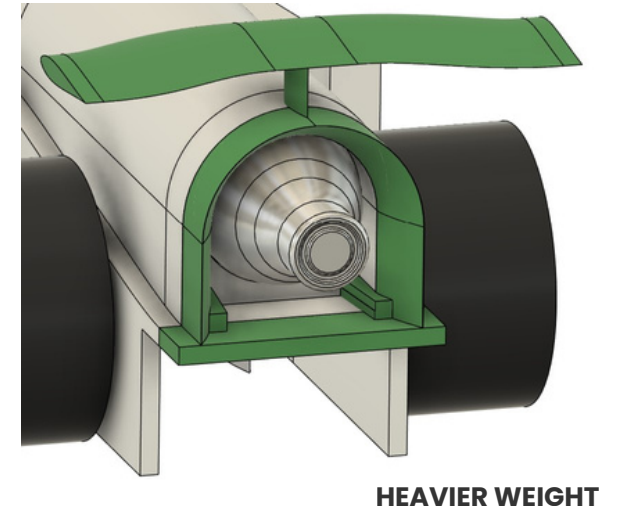
- Switched from **PLA** to **Polypropylene** (PP) for some components (1.25g/cm³ vs. 0.9g/cm³).
- Reduced wheel surface **thickness** from 2 mm to 1.5 mm.
- Replaced the inner connection with a lighter **rim-style structure**.
- Cut **excess material** from supports and bearing holders using structural optimization.
- In order not to increase the weight of the car and for reasons of time, we decided not to paint the car. We put on the stickers: name, logo, number and sponsors (plus the pirelli and f1 stickers in schools that will be given at the regional finals)

FINAL WEIGHT ESTIMATE

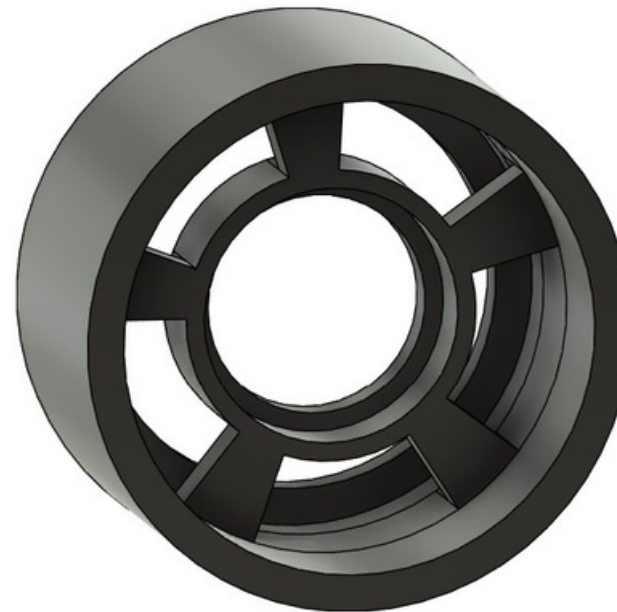
- After design optimization and printing tests, the **estimated final mass** is **≈50.2g** (without stickers).
- This is almost **45g lighter** than the qualifiers car and just **above the minimum** allowed.

PERFORMANCE RATIONALE

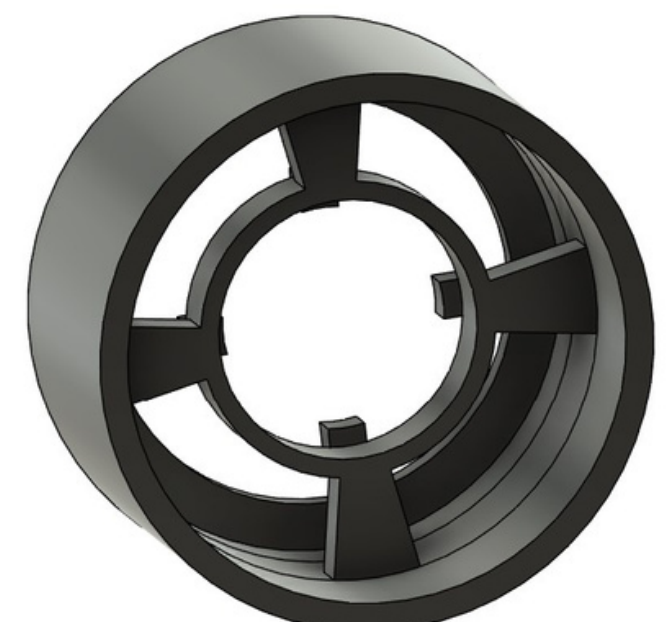
Based on our data from qualifiers, **10g** of excess weight cost us approximately **0.1 seconds** in race time — making this weight reduction crucial for improving our performance.



FIRST DESIGN OF MAIN WHEEL



FIRST WEIGHT REDUCTION



FINAL MAIN WHEEL

WEIGHT REDUCTION:

~3.4G FOR EACH WHEEL (FROM 7G TO 3.6G) AND ~44.84G FOR THE FULL CAR - W/O CAN - (FROM 95G TO 50.16G)

CFD ANALYSIS & PERFORMANCE IMPROVEMENTS

CFD TESTING PROCESS

After each major design change, we ran CFD simulations using Autodesk CFD. We focused especially on visualizing airflow patterns, pressure zones and identifying areas of turbulence. More than **10 versions of the front wing** alone were tested, together with different **rear wing supports**, various **tunnel walls and roof** specs, in order to maximise our car's performance.

KEY FINDINGS

- The final **front wing reduced drag by ~19%** compared to the initial version.
- Positioning the **Virtual Cargo behind the Halo** significantly improved the **undercut airflow**.
- Removing underbody deflectors and replacing them with **clean tunnel walls** eliminated most of the turbulence observed in the previous car.

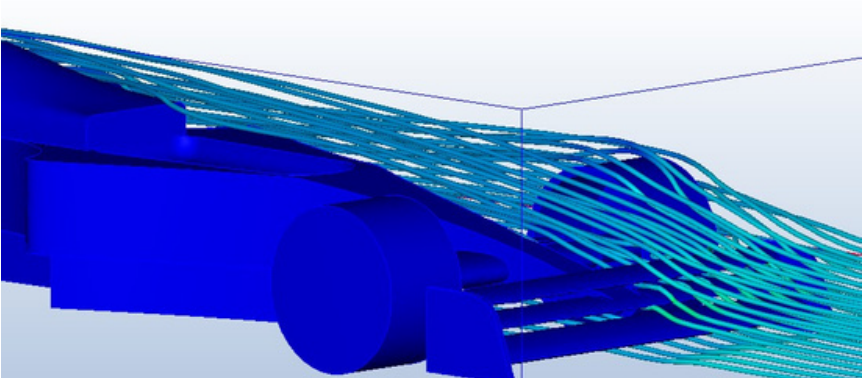
CFD TOOLS USED

- Traces to observe direction and velocity of flow
- Planes to analyze air behavior inside tunnels and around wings
- Wall calculations to identify areas of high pressure and resistance

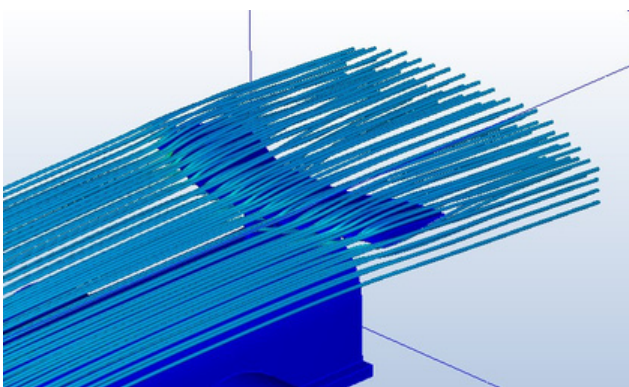
OVERALL IMPROVEMENT ESTIMATE

Our simulations suggest a total **aerodynamic improvement of ~35%** compared to the regional qualifiers' car, with major gains in drag reduction and flow cleanliness.

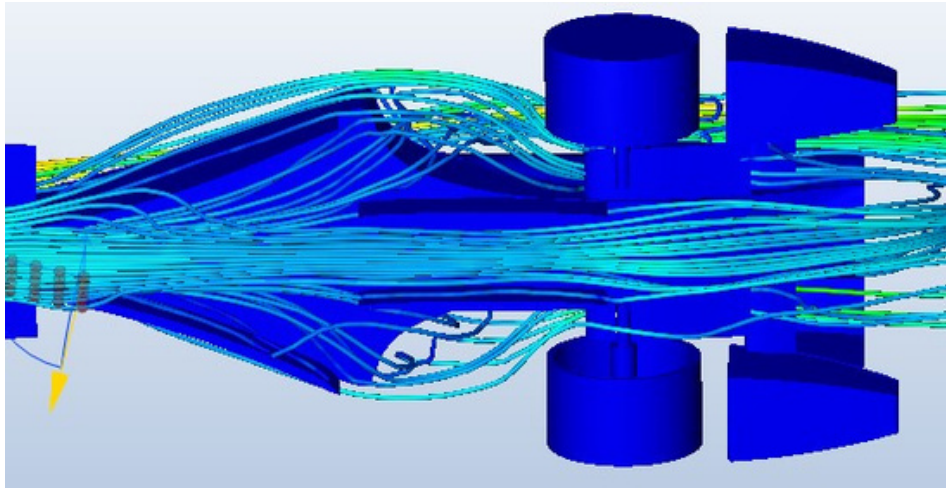
QUALIFICATIONS' CAR'S FRONT FLOW



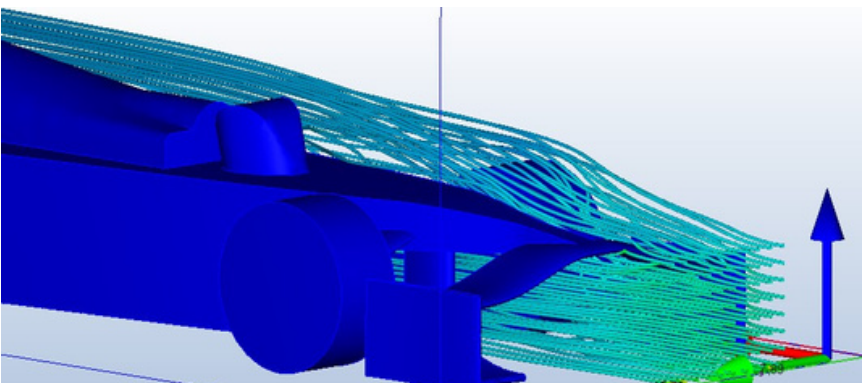
REAR WING CFD ANALYSIS: CLEAR FLOW WITHOUT TURBOLENCES



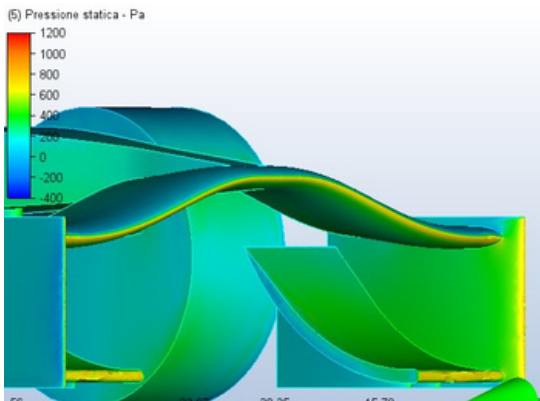
ELEMENT	RESULT
DRAG REDUCTION (FRONT)	~19%
TOTAL AERO GAIN	~35% VS QUALIFIERS CAR
FLOW TURBULENCE	SIGNIFICANTLY REDUCED



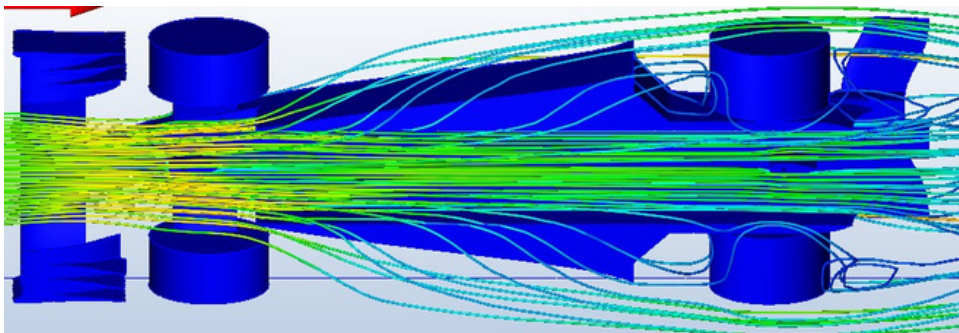
QUALIFICATIONS' CAR'S UNDERCUT FLOW: PART OF THE FLOW IS DIVERTED Laterally (COANDA EFFECT & VENTURI EFFECT) WHICH GENERATES UNWANTED TURBOLENCE



FINALS' CAR'S FRONT FLOW



FRONT WING CFD ANALYSIS



FINALS' CAR'S UNDERCUT FLOW: THE FLOW IS STRAIGHT WITHOUT UNWANTED DEVIATIONS

MATERIALS AND 3D PRINTING STRATEGY

MATERIAL SELECTION

To **reduce weight** and maintain **structural reliability**, we carefully selected materials for each printed component:

- **PLA** ($d=1.25 \text{ g/cm}^3$) was used for the front and rear wings and for the internal supports, due to its rigidity and easier printing with support materials.
- **Polypropylene** (PP) ($d=0.9 \text{ g/cm}^3$) was chosen for the remaining components because of its lower density.

However, PP is more flexible and difficult to print accurately. Some parts showed small surface irregularities, which we managed by simplifying geometries and avoiding too-thin features.

PRINTING PROCESS

- All parts were printed using FDM 3D printers, at *BITZ FabLab* (unibz): we used a *Ultimaker S3 dual extrusion* and a *Prusa XL*
- **Water-soluble** support structures were used with PLA prints to allow more complex shapes and cleaner results.
- Each component was printed at least once for testing before final assembly. As of April 29, most parts had already been approved, except for the rear wheels.

DESIGN ADJUSTMENTS

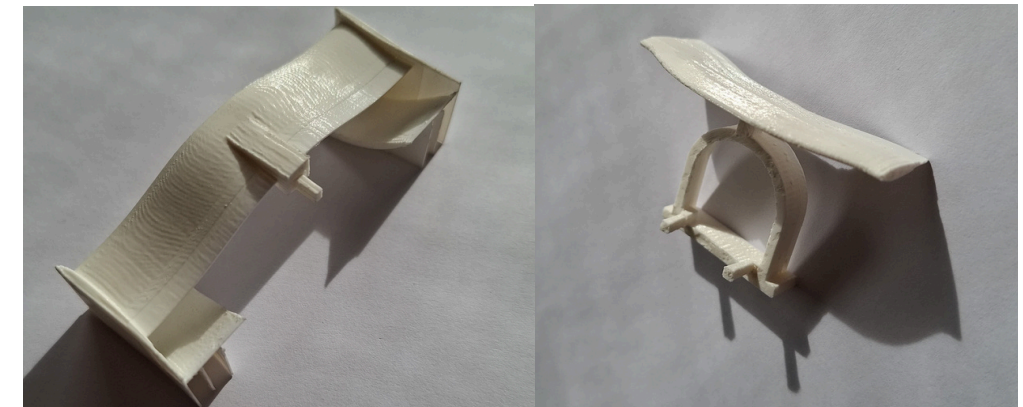
- During printing trials, we noticed **slight underweight values** in some components, especially those in PP.
- To reach the desired final weight (target: ~50g), we **reprinted** the wings and supports in PLA, slightly heavier and more rigid.

These adjustments helped balance aerodynamic efficiency, weight constraints and printability.

COMPONENT	MATERIAL	REASON
FRONT & REAR WINGS	PLA	RIGIDITY AND PRINT PRECISION
INTERNAL SUPPORTS	PLA	SLIGHTLY HEAVIER, MORE STABLE
ALL OTHER PARTS	PP	LIGHTWEIGHT, ACCEPTABLE FLEXIBILITY



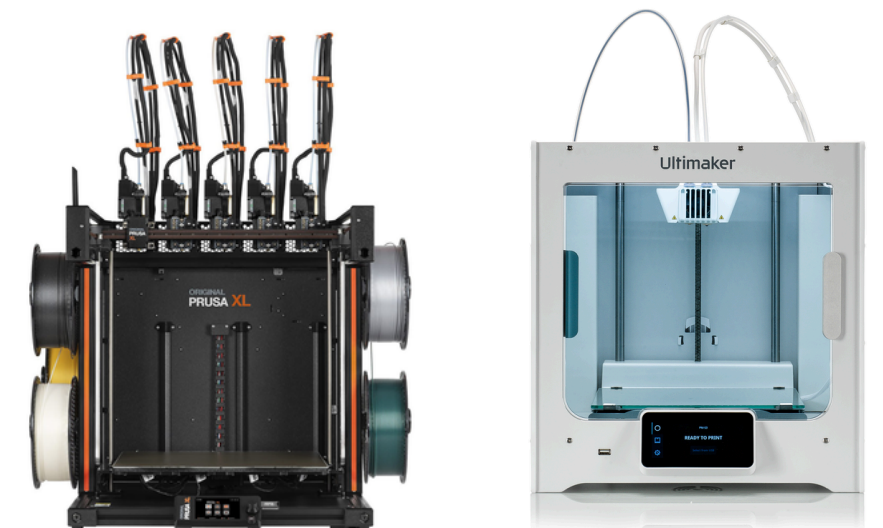
FRONT WHEEL INNER SUPPORT (IN PLA) PRINTED WITH WATER-SOLUBLE SUPPORTS



FRONT WHEEL AND REAR WING (IN WHITE PLA)



WHEEL AND WHEEL OUTER SUPPORT IN BLACK PP



THE 3D PRINTERS THAT WE USED

MANUFACTURING & ASSEMBLY

PRODUCTION STRATEGY

The manufacturing was divided between 3D printing (aerodynamic parts & structures) and milling (main body).

- **3D Printing:**
 - PLA and PP parts were printed at the local Fablab using more advanced printers. PLA parts were printed on printers with water-soluble supports (easier to remove).
 - All parts underwent print testing before approval.
- **Milling (CNC):** The body is milled from the official model block using an external provider (ziptec). Due to delays and technical limitations of local workshops, this task was concluded a bit late that expected.

ASSEMBLY APPROACH

- Our car is designed to allow **fast** and **stable** assembly.
- The wings are printed in a **single block** and connect directly to the body using integrated slots or pegs.
- Bearings are inserted into wheel supports, then aligned with dual axle pins.
- **Tolerances** were verified during early print tests.
- **Tools used:** sandpaper, files, glue.

This workflow helped us **simplify logistics**, reduce unexpected fitting problems and ensure that once the main chassis is milled, final assembly can be completed quickly and **precisely**.

TESTING AND REGULATIONS COMPLIANCE

After printing each piece, we measured it to make sure it respected the regulation. We used:

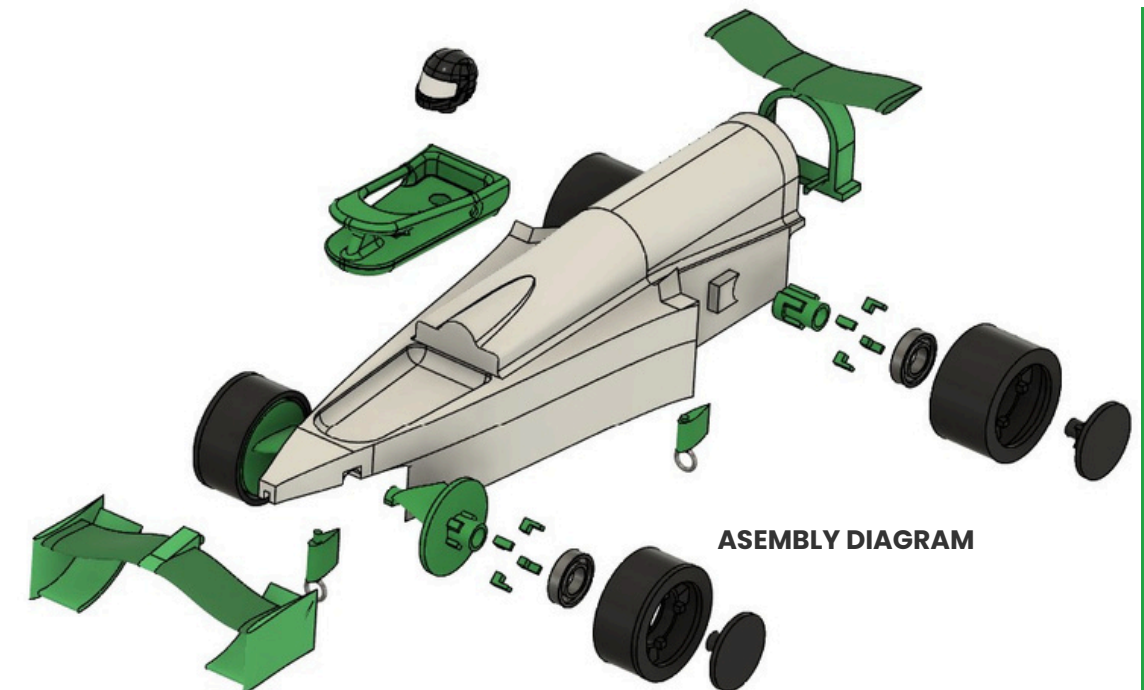
- a **digital scale** to ensure the weights corresponded to the theoretical ones and that the car didn't weight less than 50g
- a **caliber** to measure the lengths and widths of different parts
- little steel weights to test the halo's resistance

FINAL REFLECTIONS

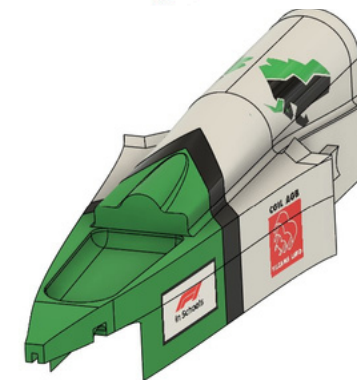
This project taught us how important it is to balance **performance**, **precision** and **planning**.

Managing time and resources was challenging and real progress only came when ideas turned into concrete action. The experience helped us grow as a team and showed us how **rewarding** it is to see a design evolve from **sketch** to **real** car.

If we had to start over, we would take **faster decisions**, communicate more clearly and leave more time for **testing** and **refinement**.



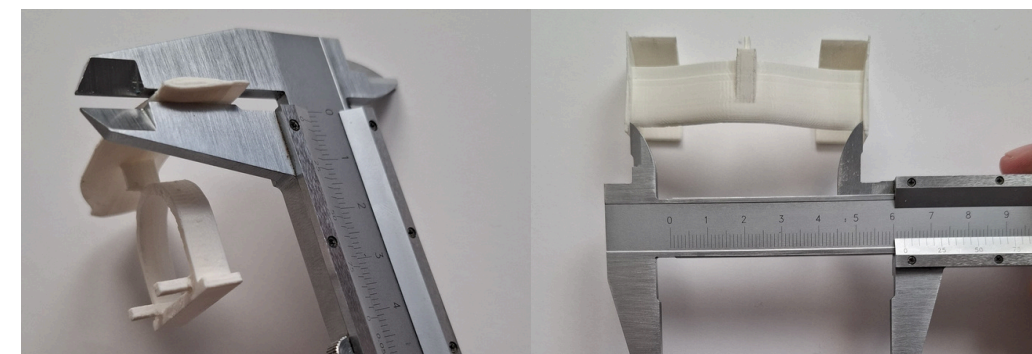
ASSEMBLY DIAGRAM



MAIN BODY MILLED BY ZIPTEC



MILLING MACHINE USED BY ZIPTEC



MEASURING FRONT WING SPAN & REAR WING THICKNESS

